

## 3.5 Conditional distributions

Throughout this section  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  is split into  $\mathbf{Y}_1 \in \mathbb{R}^{n_1}$  and  $\mathbf{Y}_2 \in \mathbb{R}^{n_2}$ , with

$$\boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix}, \quad \boldsymbol{\Sigma}_{21} = \boldsymbol{\Sigma}_{12}^\top,$$

as in Theorem 2.5.1. The question is what  $\mathbf{Y}_1$  looks like once  $\mathbf{Y}_2$  has been observed. The answer is the probabilistic core of regression.

### 3.5.1 Partial covariance with a generalized inverse

The natural answer involves  $\boldsymbol{\Sigma}_{22}^{-1}$ , but  $\boldsymbol{\Sigma}_{22}$  may be singular, for instance when  $\mathbf{Y}_2$  contains a sum and its summands. A generalized inverse (Definition 1.3.1) removes the need for a separate treatment. The facts that make this work were proved for arbitrary random vectors in [Section 2.5](#). We collect them in the notation of this chapter.

#### Lemma 3.5.1

Let  $\boldsymbol{\Sigma}$  be nonnegative definite and partitioned as above, and let  $\mathbf{G}$  be any generalized inverse of  $\boldsymbol{\Sigma}_{22}$ . Then:

- (a)  $\mathcal{C}(\boldsymbol{\Sigma}_{21}) \subseteq \mathcal{C}(\boldsymbol{\Sigma}_{22})$ , so  $\boldsymbol{\Sigma}_{21} = \boldsymbol{\Sigma}_{22}\mathbf{H}$  for some matrix  $\mathbf{H}$ ;
- (b)  $\boldsymbol{\Sigma}_{12}\mathbf{G}\boldsymbol{\Sigma}_{22} = \boldsymbol{\Sigma}_{12}$  and  $\boldsymbol{\Sigma}_{22}\mathbf{G}\boldsymbol{\Sigma}_{21} = \boldsymbol{\Sigma}_{21}$ ;
- (c)  $\boldsymbol{\Sigma}_{12}\mathbf{G}\boldsymbol{\Sigma}_{21} = \mathbf{H}^\top \boldsymbol{\Sigma}_{22}\mathbf{H}$ , which does not depend on the choice of  $\mathbf{G}$  and is nonnegative definite.

The matrix  $\boldsymbol{\Sigma}_{11.2} = \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^- \boldsymbol{\Sigma}_{21}$ , the **Schur complement** of  $\boldsymbol{\Sigma}_{22}$  in  $\boldsymbol{\Sigma}$ , is therefore well defined. It is the partial covariance matrix of  $\mathbf{Y}_1$  given  $\mathbf{Y}_2$  in the sense of Theorem 2.5.2.

*Proof.* By Theorem 3.1.2,  $\boldsymbol{\Sigma}$  is the covariance matrix of some random vector, so (a) and (b) are Theorem 2.5.1(c). For (c),  $\boldsymbol{\Sigma}_{12}\mathbf{G}\boldsymbol{\Sigma}_{21} = \mathbf{H}^\top \boldsymbol{\Sigma}_{22}\mathbf{G}\boldsymbol{\Sigma}_{22}\mathbf{H} = \mathbf{H}^\top \boldsymbol{\Sigma}_{22}\mathbf{H}$ . ■

### 3.5.2 The conditional distribution

#### Theorem 3.5.2 — Conditional distribution

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  be partitioned as above, let  $\mathbf{G}$  be a generalized inverse of  $\boldsymbol{\Sigma}_{22}$ , and put  $\mathbf{B} = \boldsymbol{\Sigma}_{12}\mathbf{G}$ . Then:

- (a)  $\mathbf{W} = \mathbf{Y}_1 - \boldsymbol{\mu}_1 - \mathbf{B}(\mathbf{Y}_2 - \boldsymbol{\mu}_2)$  is independent of  $\mathbf{Y}_2$  and  $\mathbf{W} \sim N_{n_1}(\mathbf{0}, \boldsymbol{\Sigma}_{11.2})$ ;
- (b) the conditional distribution of  $\mathbf{Y}_1$  given  $\mathbf{Y}_2 = \mathbf{y}_2$  is

$$\mathbf{Y}_1 \mid \mathbf{Y}_2 = \mathbf{y}_2 \sim N_{n_1} \left( \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} (\mathbf{y}_2 - \boldsymbol{\mu}_2), \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} \boldsymbol{\Sigma}_{21} \right), \quad (3.5.1)$$

and for  $(\mathbf{y}_2)$  in  $(\boldsymbol{\mu}_2 + \mathcal{C}(\boldsymbol{\Sigma}_{22}))$ , where  $(\mathbf{Y}_2)$  lies with one, the conditional mean does not depend on the choice of  $(\mathbf{G})$ ;

- (c) if  $\boldsymbol{\Sigma}$  is positive definite, then  $\boldsymbol{\Sigma}_{22}^{-1} = \boldsymbol{\Sigma}_{22}^{-1}$ ,  $\boldsymbol{\Sigma}_{11.2}$  is positive definite,  $\boldsymbol{\Sigma}_{11.2}^{-1}$  is the leading  $n_1 \times n_1$  block of  $\boldsymbol{\Sigma}^{-1}$ , and  $\det \boldsymbol{\Sigma} = \det \boldsymbol{\Sigma}_{22} \det \boldsymbol{\Sigma}_{11.2}$ .

*Proof.*

- (a)  $\mathbf{W} = [\mathbf{I}, -\mathbf{B}](\mathbf{Y} - \boldsymbol{\mu})$ , so  $(\mathbf{W}^\top, \mathbf{Y}_2^\top)^\top$  is an affine function of  $\mathbf{Y}$  and is jointly normal (Theorem 3.3.1). The vector  $\mathbf{W}$  is the error of the best linear predictor of  $\mathbf{Y}_1$  from  $\mathbf{Y}_2$ , so by Theorem 2.5.2(a,b) it has mean zero,  $\text{Cov}(\mathbf{W}, \mathbf{Y}_2) = \mathbf{0}$  and  $\text{Cov}(\mathbf{W}) = \boldsymbol{\Sigma}_{11.2}$ . These are moment calculations, valid for any random vector. What normality adds is that zero covariance becomes independence (Theorem 3.4.1), and that  $\mathbf{W}$ , a linear function of  $\mathbf{Y}$ , is normal.
- (b) Write  $\mathbf{Y}_1 = \boldsymbol{\mu}_1 + \mathbf{B}(\mathbf{Y}_2 - \boldsymbol{\mu}_2) + \mathbf{W}$ . The first two terms are a function of  $\mathbf{Y}_2$ , and  $\mathbf{W}$  is independent of  $\mathbf{Y}_2$ . For independent  $\mathbf{W}$  and  $\mathbf{Y}_2$ , the conditional distribution of  $h(\mathbf{W}, \mathbf{Y}_2)$  given  $\mathbf{Y}_2 = \mathbf{y}_2$  is the distribution of  $h(\mathbf{W}, \mathbf{y}_2)$  (Billingsley 1995). With  $h(\mathbf{w}, \mathbf{y}_2) = \boldsymbol{\mu}_1 + \mathbf{B}(\mathbf{y}_2 - \boldsymbol{\mu}_2) + \mathbf{w}$ , this is the normal distribution Eq. 3.5.1. If  $\mathbf{y}_2 - \boldsymbol{\mu}_2 = \boldsymbol{\Sigma}_{22} \mathbf{k}$ , then  $\mathbf{B}(\mathbf{y}_2 - \boldsymbol{\mu}_2) = \boldsymbol{\Sigma}_{12} \mathbf{G} \boldsymbol{\Sigma}_{22} \mathbf{k} = \boldsymbol{\Sigma}_{12} \mathbf{k}$ , which does not involve  $\mathbf{G}$ .
- (c) If  $\boldsymbol{\Sigma}$  is positive definite, so is its principal submatrix  $\boldsymbol{\Sigma}_{22}$ , and its only generalized inverse is its inverse. By the block determinant formula (Theorem 1.6.3), which needs only  $\boldsymbol{\Sigma}_{22}$  nonsingular,  $\det \boldsymbol{\Sigma} = \det \boldsymbol{\Sigma}_{22} \det \boldsymbol{\Sigma}_{11.2}$ . The left side and  $\det \boldsymbol{\Sigma}_{22}$  are positive, so  $\boldsymbol{\Sigma}_{11.2}$  is nonsingular. It is nonnegative definite, being  $\text{Cov}(\mathbf{W})$  by (a), hence positive definite. With both  $\boldsymbol{\Sigma}_{22}$  and its Schur complement nonsingular, the partitioned-inverse formula (Theorem 1.6.4) gives  $\boldsymbol{\Sigma}_{11.2}^{-1}$  as the leading block of  $\boldsymbol{\Sigma}^{-1}$ .

When  $\boldsymbol{\Sigma}$  is positive definite, (b) can also be read off the densities. By (c),  $\boldsymbol{\Sigma}_{11.2}$  is positive definite, so  $\mathbf{W}$  has a density. The map  $(\mathbf{y}_1, \mathbf{y}_2) \mapsto (\mathbf{w}, \mathbf{y}_2)$  has unit Jacobian, so  $f_{\mathbf{Y}}(\mathbf{y}_1, \mathbf{y}_2) = f_{\mathbf{W}}(\mathbf{y}_1 - \boldsymbol{\mu}_1 - \mathbf{B}(\mathbf{y}_2 - \boldsymbol{\mu}_2)) f_{\mathbf{Y}_2}(\mathbf{y}_2)$  by independence, and dividing by  $f_{\mathbf{Y}_2}(\mathbf{y}_2)$  gives the conditional density.

■

*Idea.* Every jointly normal vector splits as

$$\mathbf{Y}_1 = \underbrace{\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} (\mathbf{Y}_2 - \boldsymbol{\mu}_2)}_{\text{function of } \mathbf{Y}_2} + \underbrace{\mathbf{W}}_{\text{independent of } \mathbf{Y}_2}.$$

The first term is a linear regression of  $\mathbf{Y}_1$  on  $\mathbf{Y}_2$ . The second is a normal error whose distribution is the same whatever value  $\mathbf{Y}_2$  takes.

### 3.5.3 What the theorem says

Three features of [Eq. 3.5.1](#) are special to the normal distribution, and each is an assumption of the classical linear model.

- (i) *The regression is linear.*  $E(\mathbf{Y}_1 | \mathbf{Y}_2 = \mathbf{y}_2)$  is an affine function of  $\mathbf{y}_2$ , with coefficient matrix  $\Sigma_{12}\Sigma_{22}^{-1}$ .
- (ii) *The errors are homoscedastic.* The conditional covariance  $\Sigma_{11.2}$  is the same for every  $\mathbf{y}_2$ . Observing  $\mathbf{Y}_2$  tells us where  $\mathbf{Y}_1$  is centred, but not how spread out it is.
- (iii) *Conditioning cannot increase uncertainty.*  $\Sigma_{11} - \Sigma_{11.2} = \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$  is nonnegative definite ([Lemma 3.5.1\(c\)](#)), so every linear combination  $\mathbf{a}^\top \mathbf{Y}_1$  has conditional variance at most its unconditional variance. The reduction is zero iff  $\Sigma_{12} = \mathbf{0}$ .

None of these holds in general. In the mixture of Example (A mixture of two correlations), given  $Y_1 = y_1$  the variable  $Y_2$  is an equal mixture of  $N(\rho y_1, 1 - \rho^2)$  and  $N(-\rho y_1, 1 - \rho^2)$ . Its conditional mean is 0 for every  $y_1$ , but its conditional variance is  $1 - \rho^2 + \rho^2 y_1^2$ , which grows with  $|y_1|$ . Linearity and constant variance are not consequences of normal margins or of zero correlation. They are consequences of joint normality.

**One response, several predictors.** The case used most in this book has a scalar response  $Y$  and a vector  $\mathbf{X}$  of  $k$  predictors, jointly normal, with  $\text{Var}(Y) = \sigma_Y^2$ ,  $\sigma_{XY} = \text{Cov}(\mathbf{X}, Y)$  and  $\Sigma_{XX} = \text{Cov}(\mathbf{X})$  positive definite. [Theorem 3.5.2](#) says

$$Y = \alpha + \beta^\top \mathbf{X} + e, \quad \beta = \Sigma_{XX}^{-1} \sigma_{XY}, \quad \alpha = \mu_Y - \beta^\top \mu_X, \quad e \sim N(0, \sigma^2) \text{ independent of } \mathbf{X}, \quad (3.5.2)$$

with  $\sigma^2 = \sigma_Y^2 - \sigma_{XY}^\top \Sigma_{XX}^{-1} \sigma_{XY} = \sigma_Y^2(1 - \rho_{Y.X}^2)$ , where  $\rho_{Y.X}^2 = \sigma_{XY}^\top \Sigma_{XX}^{-1} \sigma_{XY} / \sigma_Y^2$  is the population squared multiple correlation ([Proposition 2.5.3](#)). So if observations  $(Y_i, \mathbf{X}_i)$  are drawn independently from a joint normal distribution, then conditionally on the predictors, the normal linear model with independent homoscedastic errors holds exactly. Inference that treats the predictors as fixed is valid for random predictors, as [Chapter 14](#) develops ([Theorem 14.1.2](#)).

**The bivariate case.** With  $k = 1$ , correlation  $\rho$  and standard deviations  $\sigma_X, \sigma_Y$ ,

$$E(Y | X = x) = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X), \quad \text{Var}(Y | X = x) = \sigma_Y^2 (1 - \rho^2).$$

In standard units,  $(E(Y | X = x) - \mu_Y) / \sigma_Y = \rho (x - \mu_X) / \sigma_X$ . The predicted response is fewer standard deviations from its mean than the predictor is from its mean, whenever  $|\rho| < 1$ . This is **regression toward the mean**, Galton's observation about the heights of parents and children (Galton 1886), and the origin of the word "regression". It is also a warning. Regressing  $X$  on  $Y$  is a different line,  $E(X | Y = y) = \mu_X + \rho(\sigma_X / \sigma_Y)(y - \mu_Y)$ , and when  $0 < |\rho| < 1$  neither line is the major axis of the density contours.

#### Example. (Two regression lines)

Take  $\mu_X = 1$ ,  $\mu_Y = 2$ ,  $\sigma_X = 1$ ,  $\sigma_Y = 1.5$  and  $\rho = 0.7$ . The regression of  $Y$  on  $X$  has slope 1.05 and residual standard deviation 1.071. Drawn in the same  $(x, y)$  plane, the regression of  $X$  on  $Y$  has slope 2.14, and the major axis of the ellipses lies between them, with slope 1.76. [Figure 3.5.1](#) shows the three lines. The line  $E(Y | X = x)$  passes through the points where each contour ellipse has a vertical tangent ([Exercise \(B2\)](#)): on a vertical line  $X = x$ , the density is highest at the conditional mean. A least squares

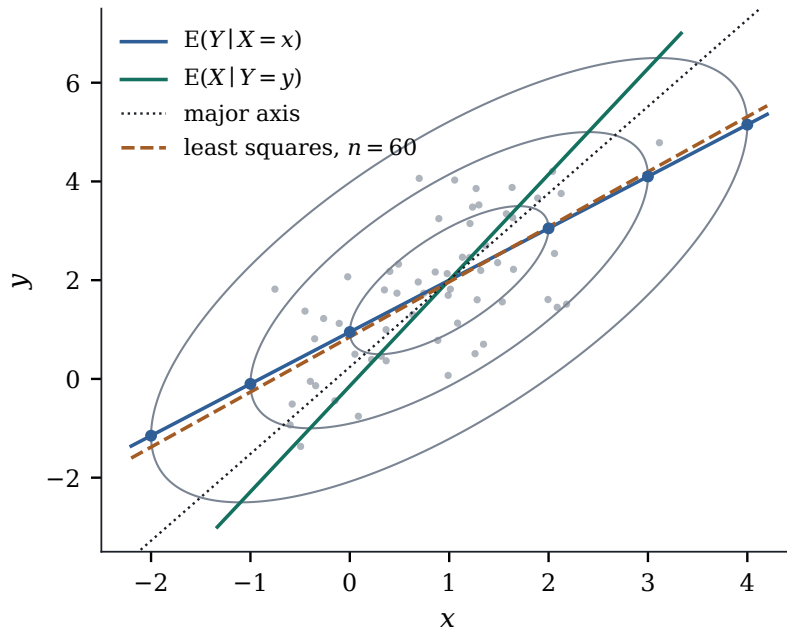


Figure 3.5.1: Contours  $\Delta^2 = 1, 4, 9$  of a bivariate normal density with  $\rho = 0.7$ . The conditional mean line  $E(Y | X = x)$  passes through the points of vertical tangency (dots). The line  $E(X | Y = y)$  passes through the points of horizontal tangency and is steeper, and the major axis lies between the two. Least squares on a sample of 60 points estimates  $E(Y | X = x)$ .

line fitted to 60 draws has slope 1.12 (standard error 0.16). It estimates the regression of  $Y$  on  $X$ , not the major axis.

### Example. (Holding a third variable fixed)

Let  $\mathbf{Y} \sim N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  with

$$\boldsymbol{\mu} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} 4 & 1 & 2 \\ 1 & 3 & 2 \\ 2 & 2 & 2 \end{pmatrix}.$$

Conditioning  $Y_1$  on  $Y_2$  alone gives slope  $\sigma_{12}/\sigma_{22} = \frac{1}{3}$  and conditional variance  $\frac{11}{3}$ . Conditioning on  $(Y_2, Y_3)$ , the listing computes  $\mathbf{B} = \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}$  in exact arithmetic and finds

$$E(Y_1 | Y_2 = y_2, Y_3 = y_3) = 3 - y_2 + 2y_3, \quad \text{Var}(Y_1 | Y_2, Y_3) = 1.$$

The coefficient of  $y_2$  changes sign. Among outcomes with the same  $Y_3$ , larger  $Y_2$  goes with smaller  $Y_1$ . Across all outcomes, larger  $Y_2$  goes with larger  $Y_1$ , because both tend to be large when  $Y_3$  is large. The conditional variance 1 also equals  $\det \boldsymbol{\Sigma} / \det \boldsymbol{\Sigma}_{22} = 2/2$ , as [Theorem 3.5.2\(c\)](#) predicts.

```
from fractions import Fraction as F
import numpy as np
rng = np.random.default_rng(305)
mx, my, sx, sy, rho = 1.0, 2.0, 1.0, 1.5, 0.7
Sigma = np.array([[sx**2, rho * sx * sy], [rho * sx * sy, sy**2]])
slope_y_on_x = rho * sy / sx          # E(Y | X = x)
slope_x_on_y = rho * sx / sy          # E(X | Y = y), as dx/dy
```

```

cond_sd = sy * np.sqrt(1 - rho**2)
n = 60
XY = rng.multivariate_normal([mx, my], Sigma, size=n)
b_ls, a_ls = np.polyfit(XY[:, 0], XY[:, 1], 1)
Linv = np.linalg.inv(np.linalg.cholesky(Sigma))
theta = np.linspace(0, 2 * np.pi, 200_001)
circle = np.vstack([np.cos(theta), np.sin(theta)])
L = np.linalg.cholesky(Sigma)
big = rng.multivariate_normal([mx, my], Sigma, size=4_000_000)
x0 = 2.0
s1 = big[np.abs(big[:, 0] - x0) < 0.01, 1]
resid = XY[:, 1] - a_ls - b_ls * XY[:, 0]
se = np.sqrt(resid @ resid / (n - 2) / np.sum((XY[:, 0] - XY[:, 0].mean())**2))
lam, U = np.linalg.eigh(Sigma)
slope_major = U[1, 1] / U[0, 1]
levels = [1, 4, 9]
gx, gy = np.meshgrid(np.linspace(-3, 5, 300), np.linspace(-4, 8, 300))
D = np.stack([gx - mx, gy - my], axis=-1)
D2 = np.einsum("...i,ij,...j->...", D, np.linalg.inv(Sigma), D)
xs = np.linspace(-2.2, 4.2, 2)
ys = np.linspace(-3.0, 7.0, 2)

mu = [F(1), F(0), F(-1)]
S = [[F(4), F(1), F(2)],
      [F(1), F(3), F(2)],
      [F(2), F(2), F(2)]]

def inv2(M):
    det = M[0][0] * M[1][1] - M[0][1] * M[1][0]
    return [[M[1][1] / det, -M[0][1] / det], [-M[1][0] / det, M[0][0] / det]]

# Y1 given (Y2, Y3): B = sigma_12 Sigma_22^{-1}, variance sigma_11 - B sigma_21
S22inv = inv2([[S[1][1], S[1][2]], [S[2][1], S[2][2]]])
s12 = [S[0][1], S[0][2]]
B = [s12[0] * S22inv[0][j] + s12[1] * S22inv[1][j] for j in range(2)]
var_1_23 = S[0][0] - (B[0] * s12[0] + B[1] * s12[1])
const = mu[0] - B[0] * mu[1] - B[1] * mu[2]
print("E(Y1 | y2, y3) =", const, "+", B[0], "y2 +", B[1], "y3; variance", var_1_23)

# (Y1, Y2) given Y3, and the partial correlation of Y1 and Y2 given Y3
C = [[S[i][j] - S[i][2] * S[2][j] / S[2][2] for j in range(2)] for i in range(2)]
print("Cov((Y1, Y2) | Y3) =", C)

```

[Run this code online](#)

### 3.5.4 The conditional mean is the best predictor

Section 6.11 defines the best linear predictor  $L(\mathbf{Y}_1 | \mathbf{Y}_2)$  as a projection, and shows that the conditional mean is the best predictor among *all* square-integrable functions of  $\mathbf{Y}_2$ . For a scalar response, the gap between the two is the approximation error of Proposition 6.11.2(c). Under joint normality the gap is zero.

### Proposition 3.5.3

Let  $\mathbf{Y}$  be jointly normal and partitioned as above, and let  $\mathbf{m}(\mathbf{Y}_2) = \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{Y}_2 - \boldsymbol{\mu}_2)$ . For every function  $\mathbf{g}$  with  $\mathbb{E}\|\mathbf{g}(\mathbf{Y}_2)\|^2 < \infty$ ,

$$\mathbb{E}\|\mathbf{Y}_1 - \mathbf{g}(\mathbf{Y}_2)\|^2 = \text{tr}(\boldsymbol{\Sigma}_{11.2}) + \mathbb{E}\|\mathbf{m}(\mathbf{Y}_2) - \mathbf{g}(\mathbf{Y}_2)\|^2.$$

So  $\mathbf{m}(\mathbf{Y}_2)$  minimizes the mean squared prediction error over all predictors, it equals the best linear predictor of Theorem 6.11.1 (entry by entry, when  $\boldsymbol{\Sigma}_{22}$  is positive definite), and the minimum is  $\text{tr}(\boldsymbol{\Sigma}_{11.2})$ .

*Proof.*  $\mathbf{Y}_1 - \mathbf{g}(\mathbf{Y}_2) = \mathbf{W} + (\mathbf{m}(\mathbf{Y}_2) - \mathbf{g}(\mathbf{Y}_2))$  with  $\mathbf{W}$  as in Theorem 3.5.2. The cross term  $\mathbb{E}[\mathbf{W}^\top(\mathbf{m}(\mathbf{Y}_2) - \mathbf{g}(\mathbf{Y}_2))]$  vanishes, because  $\mathbf{W}$  has mean zero and is independent of  $\mathbf{Y}_2$ . And  $\mathbb{E}\|\mathbf{W}\|^2 = \text{tr Cov}(\mathbf{W}) = \text{tr}(\boldsymbol{\Sigma}_{11.2})$ . The minimizer  $\mathbf{m}$  is affine, so it is also the best affine predictor, which is the best linear predictor. ■

The proof is short because independence of  $\mathbf{W}$  and  $\mathbf{Y}_2$  is much stronger than the zero correlation that defines a projection. For non-normal vectors only zero correlation is available, and the best linear predictor is merely the best *linear* predictor.

## 3.5.5 Exercises

### 3.5.5.1 A. Check your understanding

#### Exercise A1.

For  $\boldsymbol{\mu}$  and  $\boldsymbol{\Sigma}$  of Example (Holding a third variable fixed), find the conditional distribution of  $(Y_1, Y_2)$  given  $Y_3 = y_3$  and of  $Y_3$  given  $(Y_1, Y_2) = (y_1, y_2)$ . Verify that the conditional variances are smaller than the unconditional ones.

#### Exercise A2.

In the setting of Eq. 3.5.2, let  $\boldsymbol{\Omega}$  be the inverse of  $\text{Cov}((Y, \mathbf{X}^\top)^\top)$ . Show that the row of  $\boldsymbol{\Omega}$  belonging to  $Y$  is  $(1, -\boldsymbol{\beta}^\top)/\sigma^2$ , where  $\sigma^2 = \sigma_Y^2(1 - \rho_{Y.X}^2)$ . So the regression coefficients and the error variance can be read off one row of the precision matrix.

*Solution.* Put  $Y$  first and apply Exercise (B5) with the scalar block  $Y$  and  $\mathbf{Y}_2 = \mathbf{X}$ . The conditional variance is  $\omega_{YY}^{-1} = \text{Var}(Y | \mathbf{X}) = \sigma^2$ , and the coefficient vector of the conditional mean is  $-\omega_{YY}^{-1}\boldsymbol{\Omega}_{YX} = \boldsymbol{\beta}^\top$ . Hence  $\omega_{YY} = 1/\sigma^2$  and  $\boldsymbol{\Omega}_{YX} = -\boldsymbol{\beta}^\top/\sigma^2$ . By Eq. 3.5.2,  $\sigma^2 = \sigma_Y^2(1 - \rho_{Y.X}^2)$ .

### 3.5.5.2 B. Practice

#### Exercise B1.

Suppose  $\mathbf{Y}_2 \sim \text{N}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$  and that, given  $\mathbf{Y}_2 = \mathbf{y}_2$ ,  $\mathbf{Y}_1$  is  $\text{N}(\mathbf{a} + \mathbf{B}\mathbf{y}_2, \mathbf{S})$  with  $\mathbf{a}$ ,  $\mathbf{B}$  and

$\mathbf{S}$  not depending on  $\mathbf{y}_2$ . Show that  $(\mathbf{Y}_1^\top, \mathbf{Y}_2^\top)^\top$  is jointly normal and find its mean and covariance. Explain why Example (A mixture of two correlations) does not contradict this.

*Solution.* Condition on  $\mathbf{Y}_2$ :

$$\mathbb{E} e^{\mathbf{t}_1^\top \mathbf{Y}_1 + \mathbf{t}_2^\top \mathbf{Y}_2} = \mathbb{E} \left[ e^{\mathbf{t}_2^\top \mathbf{Y}_2} e^{\mathbf{t}_1^\top (\mathbf{a} + \mathbf{B}\mathbf{Y}_2) + \frac{1}{2} \mathbf{t}_1^\top \mathbf{S} \mathbf{t}_1} \right] = e^{\mathbf{t}_1^\top \mathbf{a} + \frac{1}{2} \mathbf{t}_1^\top \mathbf{S} \mathbf{t}_1} M_{\mathbf{Y}_2}(\mathbf{t}_2 + \mathbf{B}^\top \mathbf{t}_1).$$

Inserting  $M_{\mathbf{Y}_2}$  from Theorem 3.2.1, the exponent is linear plus  $\frac{1}{2} \mathbf{t}^\top \mathbf{C} \mathbf{t}$  with  $\mathbf{C} = \begin{pmatrix} \mathbf{S} + \mathbf{B}\mathbf{\Sigma}_{22}\mathbf{B}^\top & \mathbf{B}\mathbf{\Sigma}_{22} \\ \mathbf{\Sigma}_{22}\mathbf{B}^\top & \mathbf{\Sigma}_{22} \end{pmatrix}$ , so the pair is normal with mean  $((\mathbf{a} + \mathbf{B}\boldsymbol{\mu}_2)^\top, \boldsymbol{\mu}_2^\top)^\top$  and covariance  $\mathbf{C}$ . In Example (A mixture of two correlations), the conditional distribution of  $Y_2$  given  $Y_1$  is a mixture, not a normal distribution, and its variance depends on  $y_1$ .

### Exercise B2.

Show that for a bivariate normal density with  $|\rho| < 1$ , the points of each contour ellipse at which the tangent is vertical lie on the line  $y = \mathbb{E}(Y \mid X = x)$ . Explain the result in terms of the conditional density of  $Y$  given  $X = x$ .

*Solution.* Write the ellipse as  $\{\boldsymbol{\mu} + \mathbf{u} : \mathbf{u}^\top \boldsymbol{\Sigma}^{-1} \mathbf{u} = c^2\}$ . At a point with vertical tangent, the normal vector  $\boldsymbol{\Sigma}^{-1} \mathbf{u}$  is horizontal, so  $\boldsymbol{\Sigma}^{-1} \mathbf{u} = \lambda(1, 0)^\top$  and  $\mathbf{u} = \lambda(\sigma_X^2, \rho\sigma_X\sigma_Y)^\top$ . The slope  $u_2/u_1 = \rho\sigma_Y/\sigma_X$  is that of the conditional mean line, and both such points ( $\lambda > 0$  and  $\lambda < 0$ ) lie on it. Along the vertical line  $X = x$ , the joint density is proportional to the conditional density of  $Y$ , which is highest at  $\mathbb{E}(Y \mid X = x)$ . The contour that just touches the vertical line touches it at the point of highest density on that line.

### Exercise B3.

Let  $Z_1, \dots, Z_n$  be independent  $N(0, 1)$  and  $S_k = Z_1 + \dots + Z_k$ . Show that  $(S_1, \dots, S_n)$  is jointly normal with  $\text{Cov}(S_j, S_k) = \min(j, k)$ . Find the conditional distribution of  $S_j$  given  $S_n = s$ , and comment on its variance as a function of  $j$ .

*Solution.*  $(S_1, \dots, S_n)^\top = \mathbf{L}\mathbf{Z}$  with  $\mathbf{L}$  lower triangular and all entries on and below the diagonal equal to one, so the vector is normal with covariance  $\mathbf{L}\mathbf{L}^\top$ , whose  $(j, k)$  entry counts the common indices,  $\min(j, k)$ . By Theorem 3.5.2 with  $\mathbf{Y}_1 = S_j$ ,  $\mathbf{Y}_2 = S_n$ ,

$$S_j \mid S_n = s \sim N\left(\frac{j}{n}s, j - \frac{j^2}{n}\right) = N\left(\frac{j}{n}s, \frac{j(n-j)}{n}\right).$$

The conditional mean interpolates linearly between 0 and  $s$ . The variance is zero at both ends, where the value is known, and largest in the middle.

### Exercise B4.

Let  $(Y, X_1, X_2)$  be normal with positive definite covariance, and let  $\mathbf{Y}_2 = (X_1, X_2, X_1 + X_2)^\top$ . Show that  $\boldsymbol{\Sigma}_{22}$  is singular. Using a generalized inverse, show that the conditional distribution of  $Y$  given  $\mathbf{Y}_2$  from Theorem 3.5.2 coincides with its conditional distribution given  $(X_1, X_2)$ .

**Exercise B5.**

With  $\mathbf{\Omega} = \mathbf{\Sigma}^{-1}$  partitioned like  $\mathbf{\Sigma}$ , show that  $E(\mathbf{Y}_1 \mid \mathbf{Y}_2 = \mathbf{y}_2) = \boldsymbol{\mu}_1 - \mathbf{\Omega}_{11}^{-1} \mathbf{\Omega}_{12}(\mathbf{y}_2 - \boldsymbol{\mu}_2)$  and  $\text{Cov}(\mathbf{Y}_1 \mid \mathbf{Y}_2) = \mathbf{\Omega}_{11}^{-1}$ .

*Solution.* By Theorem 1.6.4,  $\mathbf{\Omega}_{11} = \mathbf{\Sigma}_{11 \cdot 2}^{-1}$  and  $\mathbf{\Omega}_{12} = -\mathbf{\Sigma}_{11 \cdot 2}^{-1} \mathbf{\Sigma}_{12} \mathbf{\Sigma}_{22}^{-1}$ . Hence  $-\mathbf{\Omega}_{11}^{-1} \mathbf{\Omega}_{12} = \mathbf{\Sigma}_{12} \mathbf{\Sigma}_{22}^{-1}$  and  $\mathbf{\Omega}_{11}^{-1} = \mathbf{\Sigma}_{11 \cdot 2}$ . Substitute into [Eq. 3.5.1](#).

**3.5.5.3 C. Going deeper****Exercise C1.**

Assume  $\mathbf{\Sigma}$  is positive definite. Prove [Theorem 3.5.2\(b\)](#) by dividing the joint density by the marginal density of  $\mathbf{Y}_2$  directly, using the partitioned inverse and the block determinant formula.